Exam-2

11. Among the following metal carbonyls, which has the lowest value of stretching frequency of CO bond?

(a) $Cr(CO)_6$ (b) $Co_2(CO)_8$

(c) $[Ti(CO)_6]^{2-}$ (d) $Mn_2(CO)_{10}$

12. Deoxy hemoglobin is

(a) six coordinated (b) five coordinated

(c) four coordinated (d) three coordinated

13. The binding affinity of the free Heme group for CO is about ---------- times than O_2

(a) 2500 (b) 0.25 lakh

(c) 2.5 lakh (d) 25 lakh

14. The saturation (with dioxygen) of the Hb in the muscle at pH = 7.6 is ----------than that of Hb in the muscle at pH = 6.8

(a) Less (b) More

(c) Equal (d) None of the above

15. The typical residence time of CO in the troposphere is about

(a) 2 weeks to 3 months (b) 3 to 13 months

(c) 2 to 3 weeks (d) 2 to 3 days

16. Which of the following is not an important parameter of soil to capture CO

(a) Water content (b) Pressure

(c) Temperature (d) None of the above

17. Which of the NOx is colored at room temperature?

(a) NO (b) N_2O_5

(c) NO_2 (d) N_2O

18. The role of Rh in the three-way catalytical converter is to convert

(a) NO to O_2 (b) CO to CO_2

(c) HC to CO_2 (d) all of the above

19. The role of Rh in the two-way catalytical converter is to convert

(a) NO to O_2 (b) None of the above

(c) HC to CO_2 (d) HC to H_2

20. What is the chemical name of $H_2S_2O_4$

(a) Thiosulphuric acid (b) Dithionic acid

(c) Dithionous acid (d) None of the above

21. Which chemical is used in the flue gas desulfurization (FGD)

(a) $Ca(CO_3)_2$ (b) $Mg(OH)_2$

(c) $BaCO_3$ (d) $Mg(NO_3)_2$

Questions (Total marks = 50)

Please tick the correct answer.

1. The typical pressure used in the Born Haber process is
   (a) 200 atm (b) 2 atm
   (c) 20 atm (d) 2000 atm
2. Which of the following is an example of green pesticides
   (a) Spinosed (b) Spinosad
   (c) Spinosat (d) None of the above
3. Which of the following is not among the five spheres of the environment:
   (a) Atmosphere (b) Biosphere
   (c) Lakosphere (d) Anthrosphere
4. The unit of absorbance is
   (a) Dimensionless (b) l/mm
   (c) mol/L (d) counts per second (cps)
5. Where is the absorption maximum located in the absorption spectrum of ozone in the 200 to 350 nm wavelength range?
   (a) 200-210 nm (b) 220-230 nm
   (c) 250-260 nm (d) 270-280 nm
6. Which of the parameters of Lambert-Beer's law is made very large for in-situ measurement of ozone in the atmosphere?
   (a) Concentration (b) Molar extinction coefficient
   (c) path length (d) Molar absorptivity
7. Which of the following is both a primary and secondary pollutant?
   (a) Ozone (b) NO_2
   (c) Lead (d) SO_2
8. How many nodes are there in the $\sigma_u^*(2p_x)$ of the CO molecule?
   (a) None (b) One
   (c) Two (d) Three
9. What is the B.O. of $^1O_2$ ($^1\Delta$)?
   (a) one (b) two
   (c) three (d) none of the above
10. The stretching frequency of bridging carbonyl is ——— than that of the terminal carbonyl in the metal carbonyl compounds
    (a) less (b) more
    (c) equal (d) none of the above

Exam-2

22. Which chemical is responsible for the brownish color of the photochemical smog?
(a) N_2O_5 (b) NO
(c) NO_2 (d) H_3C-(CO)-O-O-NO_2 (PAN)

23. One Dobson unit is equal to
(a) 2.7×10^{21} ozone/sq centimeter (b) 2.7×10^{20} ozone/sq meter
(c) 2.7×10^{19} ozone/sq meter (d) 2.7×10^{20} ozone/sq centimeter

24. Which cancer-causing metal is dumped by the Leather tanneries
(a) Cr (b) Pb
(c) Hg (d) As

25. The chemical formula of Endrin is
(a) $C_{12}H_{10}Cl_6O$ (b) $C_{12}H_8Cl_6O$
(c) $C_{12}H_{10}Cl_8O$ (d) $C_{12}H_8Cl_8O$

26. Fermentation of sugar into alcohol is a typical example of
a. Bioremediation
b. Biotransformation
c. Bioaugmentation
d. Biostimulation

27. Xenobiotic chemicals are
a. Carbon-containing compounds and biologically active
b. Not found in nature
c. Not toxic to organisms
d. Used as a growth substrate for the organisms.

28. Biostimulation is
a. Addition of xenobiotics for natural attenuation
b. Addition of nutrients for the growth of microorganisms for natural attenuation
c. Addition of microbes in supplement of existing microbes for natural attenuation
d. None of the above.

29. Bisphenol-A (BPA) is used as
a. Preservatives
b. Plasticizers
c. Dyes
d. Personal care products

30. Which one of the following is correct?
a. Cometabolism is the process of degradation of a contaminant by enzymes in microbes during the metabolism process of a growth substrate.
b. Xenobiotic compounds can be degraded by cometabolic process in microorganisms.
c. Cometabolites cannot be used as a carbon and energy source in the microorganism.
d. All the statements are correct.

45. The dirty dozen compounds do not include

a. Toluene
b. Dioxines
c. Aldrene
d. Hexacholorbenzene

46. Which is not the characteristic of a consumable product?

a. Degradable
b. Bio-accumulative
c. Non-toxic
d. Not recyclable

47. The concept of atom efficiency was invented by

a. Professor Trost
b. Professor Anastas
c. Professor J.C. Warner
d. Professor Swaminathan

48. The concepts of atom economy do not include

a. Reduction of use of non-renewable sources
b. Minimizing the byproduct formation
c. Reducing the number of steps
d. Recycling the byproduct

49. The Thalidomide tragedy shows the emergence of obeying the following rule of green chemistry

a. Less hazardous chemical synthesis
b. Designing safer chemicals
c. Safer solvent and auxiliary
d. Reduce derivatives.

50. Which of the following best describes Industrial Ecology?

a. The study of industrial systems aimed at finding strategies to reduce environmental impact for sustainability.
b. The study of forests and other natural environments to find ways to reduce and reuse natural resources.
c. An engineering specialty aimed at generating more energy output.
d. The study of turning previously infertile land into fertile land.